



3 - Predicates + Truth Tables

Week 4 – 2 Sep 2025

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Cheat sheet

Conditional Statement

$$p \rightarrow q$$

Contraposition

$$\neg q \rightarrow \neg p$$

Converse

$$q \rightarrow p$$

Inverse

$$\neg p \rightarrow \neg q$$

p only if q

$$\neg q \rightarrow \neg p$$

p iff q

$$p \leftrightarrow q$$

Cheat sheet

r sufficient condition s

$$r \rightarrow s$$

r necessary condition s

$$\neg r \rightarrow \neg s$$

r necc & suff condition s

$$r \leftrightarrow s$$

r unless s

$$\neg s \rightarrow r$$

Cheat sheet

- Commutative laws:

$$p \wedge q \equiv q \wedge p$$

$$p \vee q \equiv q \vee p$$

- Associative laws:

$$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$$

$$(p \vee q) \vee r \equiv p \vee (q \vee r)$$

- Distributive laws:

$$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$$

$$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$$

- Identity laws:

$$p \wedge \mathbf{T} \equiv p$$

$$p \vee \mathbf{F} \equiv p$$

- Negation laws:

$$p \vee \neg p \equiv \mathbf{T}$$

$$p \wedge \neg p \equiv \mathbf{F}$$

- Double negation:

$$\neg(\neg p) \equiv p$$

- Idempotent laws:

$$p \wedge p \equiv p$$

$$p \vee p \equiv p$$

- Universal bound laws:

$$p \vee \mathbf{T} \equiv \mathbf{T}$$

$$p \wedge \mathbf{F} \equiv \mathbf{F}$$

- De Morgan's laws:

$$\neg(p \vee q) \equiv \neg p \wedge \neg q$$

$$\neg(p \wedge q) \equiv \neg p \vee \neg q$$

- Absorption laws:

$$p \vee (p \wedge q) \equiv p$$

$$p \wedge (p \vee q) \equiv p$$

- Negations of **T** and **F**:

$$\neg \mathbf{T} \equiv \mathbf{F}$$

$$\neg \mathbf{F} \equiv \mathbf{T}$$

Cheat sheet

Modus Ponens	$p \rightarrow q$ p $\therefore q$	Elimination	$p \vee q$ $\neg q$ $\therefore p$	$p \vee q$ $\neg p$ $\therefore q$
Modus Tollens	$p \rightarrow q$ $\neg q$ $\therefore \neg p$	Transitivity	$p \rightarrow q$ $q \rightarrow r$ $\therefore p \rightarrow r$	
Generalization	p $\therefore p \vee q$	Proof by division into cases	$p \vee q$ $p \rightarrow r$ $q \rightarrow r$ $\therefore r$	q $\therefore p \vee q$
Specialization	$p \wedge q$ $\therefore p$			
Conjunction	p q $\therefore p \wedge q$	Contradiction Rule	$\neg p \rightarrow \mathbf{F}$ $\therefore p$	

Ex. 2.2.20e, 22e, 23e. Write the negation, contrapositive, converse, and inverse for the following statement. (Assume that all variables represent fixed quantities or entities, as appropriate.)

“If x is nonnegative, then x is positive or x is 0.”

SOLUTION . Negation: x is nonnegative and x is not positive and x is not 0.

Contrapositive: If x is not positive and x is not 0, then x is negative. (Or “... x is not nonnegative.”)

Converse: If x is positive or x is 0, then x is nonnegative.

Inverse: If x is negative, then x is not positive and x is not 0. (Or “If x is not nonnegative ...”.) □

Ex. 2.2.39. Definition: If r and s are statements, r **unless** s means if $\neg s$ then r .

Rewrite the statement in if-then form. “This door will not open unless a security code is entered.”

SOLUTION . If a security code is not entered, then this door will not open. □

Ex. 2.2.40. Rewrite the statement in if-then form. “Catching the 8:05 bus is a sufficient condition for my being on time for work.”

SOLUTION . If I catch the 8:05 bus, then I am on time for work. □

Ex. 2.2.45. Note that “a sufficient condition for s is r ” means r is a sufficient condition for s and that “a necessary condition for s is r ” means r is a necessary condition for s . Rewrite the following statement in if-then form.

A necessary condition for this computer program to be correct is that it not produce error messages during translation.

SOLUTION . Let r be “this computer program produce error messages during translation” and s be “this computer program is correct”.

Then the two acceptable forms are “if $\neg r$ then $\neg s$ ” and “if s then r ”.

If this computer program produces error messages during translation, then it is not correct.

OR

If this computer program is correct, then it does not produce error messages during translation. □

Ex. 2.2.49. (a) Use the logical equivalences

$$p \rightarrow q \equiv \neg p \vee q \quad \text{and} \quad p \leftrightarrow q \equiv (\neg p \vee q) \wedge (\neg q \vee p)$$

L1 L2

to rewrite the given statement form without the symbol $\rightarrow$ and $\leftrightarrow$, and (b) use the logical equivalence

$$p \vee q \equiv \neg(\neg p \wedge \neg q) : \text{Replace all the } \mathbf{V}$$

to rewrite the given statement form using only $\wedge$ and $\neg$.

Given statement: $(p \rightarrow r) \leftrightarrow (q \rightarrow r).$

Ex. 2.2.49. (a) Use the logical equivalences

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to rewrite the given statement form without the symbol $\rightarrow$ and $\leftrightarrow$, and (b) use the logical equivalence

$$p \vee q \equiv \neg(\neg p \wedge \neg q)$$

to rewrite the given statement form using only $\wedge$ and $\neg$.

Given statement: $(p \rightarrow r) \leftrightarrow (q \rightarrow r)$.

SOLUTION . (a)

$$\begin{aligned} & (p \rightarrow r) \leftrightarrow (q \rightarrow r) \\ & \equiv (\neg p \vee r) \leftrightarrow (\neg q \vee r) && \text{(by first logical equivalence in (a))} \\ & \equiv [\neg(\neg p \vee r) \vee (\neg q \vee r)] \wedge [\neg(\neg q \vee r) \vee (\neg p \vee r)] && \text{(by second logical equivalence in (a))} \\ & && \text{(an acceptable answer)} \\ & \equiv [(p \wedge \neg r) \vee (\neg q \vee r)] \wedge [(q \wedge \neg r) \vee (\neg p \vee r)] && \text{(by De Morgan's law)} \\ & && \text{(another acceptable answer)} \end{aligned}$$

(b) Continuing from Part (a),

$$\begin{aligned} & \equiv [(p \wedge \neg r) \vee \neg(\neg\{\neg q\} \wedge \neg r)] \wedge [(q \wedge \neg r) \vee \neg(\neg\{\neg p\} \wedge \neg r)] && \text{(by logical equivalence in (b))} \\ & \equiv [(p \wedge \neg r) \vee \neg(q \wedge \neg r)] \wedge [(q \wedge \neg r) \vee \neg(p \wedge \neg r)] && \text{(by double negation)} \\ & \equiv \neg[\neg(p \wedge \neg r) \wedge \neg\{\neg(q \wedge \neg r)\}] \wedge \neg[\neg(q \wedge \neg r) \wedge \neg\{\neg(p \wedge \neg r)\}] && \text{(by logical equivalence in (b))} \\ & \equiv \neg[\neg(p \wedge \neg r) \wedge (q \wedge \neg r)] \wedge \neg[\neg(q \wedge \neg r) \wedge (p \wedge \neg r)] && \text{(by double negation)} \end{aligned}$$

□

Ex. 2.3.11. Use truth tables to determine whether the argument form is valid. Indicate which columns represent the premises and which represent the conclusion, and include a few words of explanation to support your answers.

$$\begin{aligned} & p \rightarrow r \\ & q \rightarrow r \\ \therefore & p \vee q \rightarrow r. \end{aligned}$$

SOLUTION .

				premises		conclusion	
p	q	r	$p \vee q$	$p \rightarrow r$	$q \rightarrow r$	$p \vee q \rightarrow r$	
1	1	1	1	1	1	1	← critical row
1	1	0	1	0	0		
1	0	1	1	1	1	1	← critical row
1	0	0	1	0	1		
0	1	1	1	1	1	1	← critical row
0	1	0	1	1	0		
0	0	1	0	1	1	1	← critical row
0	0	0	0	1	1	1	← critical row

The truth table shows that in every situation (represented by rows 1, 3, 5, 7, and 8) in which all the premises are true, the conclusion is also true. Therefore, the argument is valid. $\square$

Critical rows are rows in which **ALL the premises are true**

Ex. 2.3.12. Use truth tables to show that the following forms of argument are invalid.

a. $p \rightarrow q$
 q
 $\therefore p$
 (converse error)

b. $p \rightarrow q$
 $\neg p$
 $\therefore \neg q$
 (inverse error)

SOLUTION . (a)

		premises		conclusion	
p	q	$p \rightarrow q$	q	p	
1	1	1	1	1	← critical row
1	0	0	0		
0	1	1	1	0	← critical row
0	0	1	0		

Row 3 of the truth table shows that it is possible for an argument of this form to have true premises and a false conclusion. Hence the argument form (inverse error) is invalid.

Ex. 2.3.12. Use truth tables to show that the following forms of argument are invalid.

a. $p \rightarrow q$
 q
 $\therefore p$
 (converse error)

b. $p \rightarrow q$
 $\neg p$
 $\therefore \neg q$
 (inverse error)

(b)

		premises		conclusion	
p	q	$p \rightarrow q$	$\neg p$	$\neg q$	
1	1	1	0		
1	0	0	0		
0	1	1	1	0	← critical row
0	0	1	1	1	← critical row

Row 3 of the truth table shows that it is possible for an argument of this form to have true premises and a false conclusion. Hence the argument form (inverse error) is invalid. □

SKIP

Ex. 2.3.40. Sharky, a leader of the underworld, was killed by one of his own band of four henchmen. Detective Sharp interviewed the men and determined that all were lying except for one. He deduced who killed Sharky on the basis of the following statements:

- a. Socko: Lefty killed Sharky.
- b. Fats: Muscles didn't kill Sharky.
- c. Lefty: Muscles was shooting craps with Socko when Sharky was knocked off.
- d. Muscles: Lefty didn't kill Sharky.

Who did kill Sharky?

SOLUTION . First, observe that Socko and Muscles are saying opposite things, so one of them must be telling the truth.

Suppose Socko is telling the truth.

Then Lefty killed Sharky.

(That is what Socko said.)

Then Fats must be lying.

(All were lying except one.)

Then the statement "Muscles didn't kill Sharky" is false.

(Since Fats is lying.)

Therefore Muscles killed Sharky.

(By negation.)

Therefore both Muscles and Lefty killed Sharky.

(Contradiction to only one henchmen killed Sharky.)

Ex. 2.3.43. Use the valid argument forms listed in the lecture notes to deduce the conclusion from the premises, giving a reason for each step. Assume all variables are statement variables.

- a. $\neg p \rightarrow r \wedge \neg s$
- b. $t \rightarrow s$
- c. $u \rightarrow \neg p$
- d. $\neg w$
- e. $u \vee w$
- f. $\therefore \neg t$

Modus Ponens	$p \rightarrow q$ p $\therefore q$	Elimination	$p \vee q$ $\neg q$ $\therefore p$	$p \vee q$ $\neg p$ $\therefore q$
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Generalization	p $\therefore p \vee q$	Proof by division into cases	$p \vee q$ $p \rightarrow r$ $q \rightarrow r$ $\therefore r$	
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Conjunction	$p \wedge q$ p q $\therefore p \wedge q$	Contradiction Rule	$\neg p \rightarrow \mathbf{F}$ $\therefore p$	

Ex. 3.1.7. Find the truth set of each predicate.

- a. predicate: $6/d$ is an integer, domain: $\mathbb{Z}$.
- b. predicate: $6/d$ is an integer, domain: $\mathbb{Z}^+$.
- c. predicate: $1 \leq x^2 \leq 4$, domain: $\mathbb{R}$.
- d. predicate: $1 \leq x^2 \leq 4$, domain: $\mathbb{Z}$.

Bonus: *predicate* : $d_1/d_2, d_1 \in \mathbb{Z}, d_2 \in \mathbb{N}$

SOLUTION . a. $6/d$ is an integer when d is any of the integers $-6, -3, -2, -1, 1, 2, 3$, and 6 . Therefore the truth set is $\{-6, -3, -2, -1, 1, 2, 3, 6\}$.

b. The truth set is $\{1, 2, 3, 6\}$.

c. Note that $1 \leq x^2 \leq 4$ is equivalent to two inequalities,

$$\begin{array}{lll} 1 \leq x^2 & \text{and} & x^2 \leq 4. \\ \therefore x \geq 1 \text{ or } x \leq -1 & \text{and} & -2 \leq x \leq 2 \\ \therefore x \geq 1 \text{ and } -2 \leq x \leq 2 & \text{or} & x \leq -1 \text{ and } -2 \leq x \leq 2 \\ \therefore 1 \leq x \leq 2 & \text{or} & -2 \leq x \leq -1. \end{array}$$

Therefore, the truth set is $\{x \in \mathbb{R} \mid -2 \leq x \leq -1 \text{ or } 1 \leq x \leq 2\}$.

d. The truth set is $\{x \in \mathbb{Z} \mid -2 \leq x \leq -1 \text{ or } 1 \leq x \leq 2\} = \{-2, -1, 1, 2\}$.



Extra Qn: AY22/23 S1 Midterm

$$(\neg p \wedge q) \rightarrow r$$

$$\neg p$$

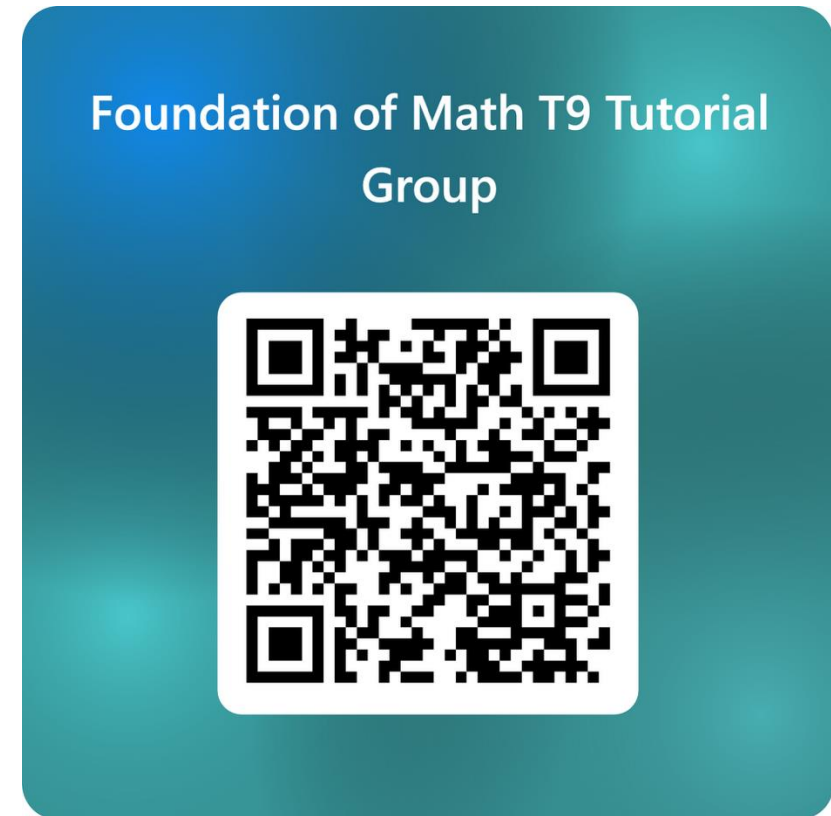
$$\neg(p \vee r)$$

$$\therefore \neg q$$

Introduction

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